METRIC
MIL-T-70988 (AR)
AMENDMENT 1
06 JANUARY 1992

MILITARY SPECIFICATION

TRACER ASSEMBLY, 12525043

This Amendment forms a part of Military Specification MIL-T-70988 (AR), dated 15 August 1990, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	Reprinted without change
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* Indicates specific pages revised by this Amendment.

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.5	TITLE Cup, Tracer		SHEET 1 OF 1		DRAWING NUMBER 12525039
					NEXT HIGHER ASSEMBLY 12525043
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Length		0.40%	3.2	Gage
103	Primer coating missing		0.40%	3.2	Visual
104	Threads in Zone A		0.40%	3.2	Gage
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Thickness through bottom		0.65%	3.2	Gage
203	Burrs		0.65%	3.5	Visual
204	Foreign matter		0.65%	3.5	Visual
205	Crack, puncture, or other damage		0.65%	3.5	Visual

NOTES:

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Supersedes page 12 dated 5 August 1980.

QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH *4.4.2.6	TITLE Tracer Assembly		SHEET 1 OF 1		DRAWING NUMBER 12525043
					NEXT HIGHER ASSEMBLY 12525001
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, maximum		0.40%	3.2	Gage
102	Total Length		0.40%	3.2	Gage
103	Charge missing or less than minimum, required depth from shoulder		0.40%	3.2	Gage
104	Igniter charge chipped or broken on any one step in excess of 1/3 of the total area of the step, or total aggregate of chipped or broken area of all steps in excess of 1/3 of the total area of the largest step		0.40%	3.5	Visual
105	Static burn time	n = 10 c = 0	N/A	3.3.1.1	4.5.1.1
106	Operational Test				
	Tracer Function (21 °C ± 1.5 °C)	n = 10 c = 1	N/A	3.3.2.1	4.5.1.2
	Tracer Function (-32 °C + 3.0 °C)	n = 10	N/A		4.5.1.2
	Tracer Function (+52 °C - 3.0 °C)	n = 10	N/A		4.5.1.2

NOTES:
n = Number of samples
c = Number of allowable failures, additional failures are cause for rejection of the lot.

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4.4.3 Inspection equipment. - The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of Inspection.

4.5.1 Tracer performance.

4.5.1.1 Static performance. - The tracer assembly shall be loaded into fins or simulator mounts with the same process used in production of fin and tracer assemblies as specified in Drawing 12525001. Test units shall then be secured in a vertical or horizontal position. A quickmatch, M100 electric match or equivalent, shall be attached to the closure plug in such a manner as to ignite the igniter composition. The electric match shall then be ignited remotely and test instrumentation observed to verify proper tracer static function. In case of failure of the igniter compound, the ignition process may be repeated. Test equipment shall be in accordance with 4.4.3.

4.5.1.2 Operational performance. - The tracer assembly shall be loaded into fins using the same processes as apply in production per drawing 12525000 and 12525001. The appropriate number of test cartridges shall be assembled.

4.5.1.2.1 Testing. - A total of 30 cartridges shall be tested and evaluated for tracer performance. The 30 cartridges shall consist of ten cartridges each conditioned to 21 ± 1.5 °C, 52 - 3 °C and -32 + 3 °C. The test cartridges shall be temperature conditioned for 24 hours minimum. All cartridges shall be fired from a M256 gun.

4.5.1.2.2 Tracer function. - Tracer performance of each cartridge shall be monitored for conformance to the tracer function requirements as follows:

a. All tests requiring verification of proper tracer function shall be performed during daylight hours.

* b. Verification of proper tracer function shall be performed visually by three observers located behind the gun (see 6.9). All three observers shall utilize binoculars with 7 x by 50 mm magnification. A tracer shall be considered to have met requirements if at least one observer confirms that the tracer functioned properly.

* c. Deleted

d. A video camera shall be located at the line of fire and record tracer functioning from approximately 250 meters to 750 meters. In addition a video camera shall be located to record proper functioning at 750 meters. The videotape recordings from these cameras shall be utilized, whenever necessary, to assist the observers in their decision.

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e. A video camera shall be located downrange so as to record tracer functioning at the 2500 meter target. This recording shall be utilized for informational purposes only. All videotape recordings shall be analyzed to determine the rate of tracer functioning at 2500 meters. This information shall be provided to the technical agency to support a government data base on tracer functioning.

4.5.1.2.3 Tracer performance shall be considered acceptable if no more than one tracer fails to meet the tracer functioning requirements specified in paragraph 3. If more than one tracer fails to meet requirements, the lot shall be referred to the technical agency for disposition. Performance of tracers fired at -32 °C and +52 °C will be noted but shall not be considered for acceptance of the lot.

4.5.2 Moisture and total volatile content of tracer and igniter charge. - The contractor shall provide adequate controls to ensure the compliance of each lot with the applicable requirement. The contractor shall test a minimum of one randomly selected sample from each batch or from each 8 hours production, whichever is the lesser amount. Composite samples shall not be used for determining compliance. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity represented by the sample. The test shall be performed in accordance with MIL-STD-1234 Method 102.2.1.

4.5.3 Density of tracer and igniter charge. - The density of the tracer and igniter charge pellets shall be determined by the following method. The contractor shall calculate density on a minimum of 10 randomly selected samples from each batch run, or from each 8 hours of production. Each pellet shall be weighed and measured for length and diameter. Volume and density shall then be calculated using these measurements. Tracer pellet density shall be per drawing 12525043. Igniter pellet density shall be per drawing 12525045. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity represented by the sample.

5. PACKAGING

5.1 Preservation and packaging.

A. The tracer assembly, 12525043 shall be preserved and packaged to comply with the minimum requirements of (DOT) CFR 49, Part Number 173.105, entitled, Percussion, Tracer, Combination, Time Fuzes and Tracers. Preservation per Section 5.2.1, this specification.

5.2 Packing.

A. Tracer assembly shall be packed to provide:

B. Protection against normal transportation/vibration shock (Federal Test Method Standard 101, Method 5019.1).

C. Protection against internally/externally imposed stacking pressures.

D. Relative humidity protection during shipping/storage (10 percent R.H. until assembled).

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E. Strong, tight, outside wooden or DOT Specification 23F Shipping Container, per CFR 49, Part Number 173.105.

5.3 Marking.

A. Marking of inner and outer containers to be per MIL-STD-129 and CFR 49, Part Number 173.105.

B. Labeling of shipping container per CFR 49, Part Number 173.105, and MIL-STD-129.

5.4 Data card. - Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Intended use. - The items covered by this specification are intended for use in the 120 mm ammunition.

6.2 Ordering data. - See MIL-A-48078.

6.2.1 Acquisition requirements. - See MIL-A-48078.

6.2.2 Data requirements. - When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, subpart 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph	Data Requirement	Applicable DID No.
4.4.3	Inspection Equipment Design Approval	DI-R-1714
6.2.3	Ammunition Data Cards	DI-MISC-80043

(Data Item Descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of Data Item Descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 Submission of test data and ammunition data cards. - In addition to the normal distribution of records when the projectile is procured by the U.S. Army Armament, Materiel and Chemical Command, one copy of all ballistic tests data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: SMCAR-CCH-V, and AMSMC-QAT-A(D), Picatinny Arsenal, New Jersey 07806-5000; and Commander, AMCCOM, ATTN: AMSMC-QAD(R), Rock Island, Illinois, 61299-6000.

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6.3 Tracer definition. - The tracer as referred to in this specification consists of two tracer charge pellets and a igniter pellet compressed in a tracer cup.

6.4 Moisture. - The igniter and tracer compositions, and the pellets formed thereof, are hygroscopic and will deteriorate rapidly when exposed to moisture. Small amounts of moisture present in or absorbed by the compositions or pellets, will eventually result in unsatisfactory functioning. It is highly desirable that the ingredients, compositions, and pellets be carefully safeguarded from exposure to moisture and humid atmospheres. For insurance of a satisfactory tracer function, it is recommended that the blending and pelleting buildings be air conditioned in a manner which will hold the humidity of the atmosphere less than 60 percent.

6.5 Alternate testing procedures. - Alternative testing procedures may be proposed by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate that their effectiveness is equal to or better than the quality assurance provisions specified herein. Alternative testing procedures must have the approval of the Government Contracting Officer prior to use.

6.6 Shipping. - If shipment is necessary, Department of Transportation (DOT) Class and Marking cited on the affected item drawings is applicable. Material must be adequately and properly packaged in accordance with applicable DOT Packaging Requirements (paragraph 173.61, CODE OF FEDERAL REGULATIONS (CFR) Title 49).

6.7 Hazard notice. - The tracer and igniter charge and the tracer and igniter charge composition described herein are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.8 Drawings. - Drawings listed in Section 2 of this specification in the heading U.S. Army Armament Research and Development Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament and Development Center drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

* 6.9 Location of tracer observers. - Location of the tracer observers is advisory. However, be advised that for cartridge lot acceptance, the verification of proper tracer function shall be performed by observers located approximately thirty (30) meters off

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line, approximately fifty (50) meters behind the gun at approximately muzzle level.

* 6.10 Subject term (keyword) listing. -
Ammunition
Ballistic Testing
Cannon (120 mm)
Cartridge (120 mm)
Tank
Tracer

Custodian
Army (AR)

Preparing activity
Army (AR)

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